vitae

Abiel Mendes dos Santos

Software Engineering • Infrastructure • DevOps
Linux | Windows Server | Docker | K8S | Ansible | Proxmox | AWS | Cloudflare | MikroTik | Zabbix | Prometheus
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Summary

Engineering professional in training, studying Computer Engineering and a Network Technician degree at IFMA, with a degree in Network and Systems Administration from SENAC and 2 years of complementary training at INCODE Tech School, through Oxygeni HUB at CEUMA University.

Worked at the Public Archive of the State of Maranhão (Arquivo Público do Estado do Maranhão), supporting technical support, IT infrastructure maintenance, network administration, document organization, archive digitization, and internal departments. Organized, communicative, analytical, and proactive, with a fast learning curve for new technologies, strong problem-solving skills, and experience collaborating in multidisciplinary teams.

Currently developing skills in IT infrastructure, computer networks, and software development and operations (DevOps), while running personal infrastructure projects through Grinmaldis and offering technical support and computer maintenance services through Fornalha Tech, focused on mid- and high-end hardware.

Professional Experience

Self-employed - End-User IT Services

Fornalha TECH – São Luís, MA
08/2025 – Present

I provide IT services focused on support, maintenance, and optimization of computing environments for home and professional users. I deliver tailored solutions involving technical diagnosis, hardware maintenance, system configuration, performance improvement, and guidance on purchasing and upgrading equipment.

Activities and Achievements:

- Log, follow up, and update tickets in the GLPI ITSM platform, keeping the service lifecycle controlled and each solution documented.
- Provide specialized technical support for desktops and laptops, including diagnosis, troubleshooting, and resolution of hardware, software, and firmware (BIOS/UEFI) incidents.
- Perform preventive and corrective maintenance on desktops and laptops, including technical cleaning, component replacement, upgrades, performance tuning, and hardware and software compatibility checks.
- Install, configure, and recover Windows operating systems, including data migration, backup procedures, and file recovery.
- Provide remote and on-site support, helping users with device setup, fault resolution, technical questions, and good usage practices.
- Analyze performance and recommend upgrades and equipment purchases, weighing upgrade scenarios, cost-benefit, and usage requirements for home and professional users.
- Work with equipment across performance levels, including workstations and high-end computers, applying technical criteria for assembly, compatibility, and system stability.

Technologies and Tools: Windows, Linux, BIOS/UEFI, LAN/WAN networks, TCP/IP, FTTH, GLPI, Wireshark, Nmap, Advanced IP Scanner, Sysinternals Suite, Windows Terminal, Event Viewer, PowerShell, DISM/SFC, HWiNFO, CPU/GPU-Z, CrystalDiskMark/Info, MemTest86, OCCT, FurMark/3DMark, NVIDIA MATS/MODS, Recuva, Macrium Reflect, Rufus, Ventoy, and other connectivity, traffic analysis, maintenance, and system recovery diagnostic tools.

Intern - Administrative and IT Support

Arquivo Público do Estado do Maranhão – São Luís, MA
11/2024 – 05/2026

I supported the institution’s administrative and technology activities, helping maintain IT infrastructure, organize documents, digitize the archive, and support internal departments. I worked on network administration, technical support, infrastructure documentation, asset management, and administrative processes.

Activities and Achievements:

- Administered and maintained network infrastructure components, including a MikroTik RB750Gr3, a TP-Link TL-SG2428P switch, and TP-Link EAP610 access points.
- Supported the deployment and maintenance of structured cabling, switches, routers, access points, LAN/Wi-Fi connectivity, and the physical organization of the infrastructure.

- Provided technical support to users, performing preventive and corrective maintenance on computers and resolving hardware, software, and network incidents.
- Documented and organized the IT infrastructure using GLPI, NetBox, BookStack, and diagrams.net, keeping records of assets, services, and environment topology.
- Managed the IT asset inventory, maintenance records, and technical documentation, improving the organization and traceability of technology resources.
- Supported the administration and monitoring of two 500 Mbps Internet links, contributing to the availability and continuity of network services.
- Supported the digitization and computerization of the document archive, keeping the equipment and systems used in the process running.
- Carried out document digitization, organization, cataloging, indexing, and search, as well as drafting and reviewing administrative documents using Microsoft Office.

Technologies and Tools: MikroTik RouterOS, TP-Link Omada, TP-Link JetStream, TCP/IP, LAN/Wi-Fi, structured cabling, Windows, Linux, Microsoft Office (Word, Excel), GLPI, NetBox, BookStack, Git, diagrams.net, and digitization and document management tools.

STAFF, Monitor - Programming Marathon (Hackathon)

INCODE Tech School - Trainee – São Luís, MA
09/2024 – 09/2024

Activities and Achievements:

- Followed the teams during the challenges, providing technical support and helping resolve problems without interrupting the activities.
- Ensured the event's rules and obligations were followed, keeping the competition fair and organized.
- Supported logistics and the organization of the venue, helping the event run with fewer delays.
- Helped keep the environment safe and welcoming for participants.

Projects

Homelab – Network Infrastructure for Experimentation and Development

Infrastructure Administrator, Personal Project, 2023 – Present

I design and run a personal infrastructure for studying and practicing systems administration, networking, and DevOps. The environment uses Linux servers (Debian, Ubuntu, and CentOS) for virtualization, automation, security, observability, and service hosting experiments.

Activities and Achievements:

- Built CI/CD pipelines with GitHub Actions and GitLab CI for automated deployment and continuous validation.
- Administered containerized environments with Docker and Kubernetes clusters to orchestrate internal and external services.
- Automated infrastructure provisioning and management with Ansible, Puppet, and Terraform.
- Applied security practices, including secrets management with HashiCorp Vault and self-hosted Bitwarden, multi-factor authentication, and Zero Trust concepts.
- Implemented reverse proxy and service publishing with NGINX, Traefik, and Cloudflare.
- Set up monitoring and observability with Prometheus, Grafana, and Zabbix, including alerts and automated integrations.
- Administered Linux environments, applying hardening practices and managing networks and services with pfSense and OpenWRT.
- Documented the infrastructure using GLPI, NetBox, BookStack, diagrams.net, and Git, keeping records of assets, services, and topology.

Technologies and Tools: Linux, Docker, Kubernetes, KVM, Proxmox, VMware, OpenStack, Ansible, Terraform, Git, GitHub Actions, GitLab CI/CD, Jenkins, NGINX, Traefik, Cloudflare, pfSense, OpenWRT, Prometheus, Grafana, Zabbix, HashiCorp Vault, Bitwarden, GLPI, NetBox, BookStack, and diagrams.net.

Vrykolakas – Community and Minecraft Server Administration

Server Administrator and Developer, Personal Project, 2024 – Present

I administer and contribute to Vrykolakas, a Minecraft community focused on immersive experiences with custom servers. I use Linux servers (Ubuntu, Debian) and tools such as Docker, Bash scripting, and plugins to manage and tune the game infrastructure, aiming for high availability and performance.

Activities and Achievements:

- Set up and maintain Minecraft servers on Linux, using Docker to isolate environments and optimize resources.
- Researched and implemented automation with Bash scripts for administrative tasks such as automatic backups and scheduled restarts, reducing manual work and making processes more predictable.
- Managed plugins (e.g., Spigot, Paper, LuckPerms) for gameplay customization, user permissions, and protection against griefing.

- Configured reverse proxy with NGINX and integration with Cloudflare, improving latency and protecting against DDoS attacks.
- Integrated monitoring systems such as Prometheus and Grafana to track server metrics such as CPU, memory, and player connections.
- Implemented a Discord notification system using webhooks, informing the community about updates, events, and maintenance alerts.
- Maintain the infrastructure on Linux servers, applying hardening policies based on CIS guides, and plan to use pfSense for firewall management.
- Documenting the server setup and community processes on our site, including guides for installing mods and administering plugins.

Technologies and Tools: Linux (Debian, Ubuntu), Docker, Bash, Java, Minecraft Servers, Forge, PaperMC, Spigot, LuckPerms, Cloudflare, NGINX, Git, GitHub, GitHub Actions, Prometheus, Grafana, Zabbix, Discord Webhooks, PostgreSQL/MySQL, Spark Profiler, OpenJDK, Linux systemd, SSH, and automated backups.

Education

Technician in Network and Systems Administration, *SENAC*
Centro, São Luís, MA
03/2025 – 08/2025

Technician in Computer Networks, *Federal Institute of Maranhão (IFMA)*
São José de Ribamar, MA
03/2024 – Present

Module 2 - Oxygenation, Software Engineering, *INCODE Tech School/CEUMA*
Turu, São Luís, MA
08/2024 – 02/2025

Module 1 - Foundation, Software Engineering, *INCODE Tech School/CEUMA*
Turu, São Luís, MA
02/2024 – 07/2024

High School, Robotics, FLL/FIRA, CEM Almirante Tamandaré
COHAB, São Luís, MA
01/2020 – 01/2023

Skills

- Development: Python, Java, TypeScript, Node.js, REST APIs, Git, GitHub, GitLab, Jenkins, Gitea
- Infrastructure: Docker, Proxmox, VMware, KVM, Kubernetes, OpenStack, EVE-NG, XCP-ng, pfSense, OpenWRT
- Automation and Configuration: Ansible, Terraform
- Monitoring and Logging: Zabbix, Prometheus, Grafana
- Databases: MySQL, Firebase, H2
- Networking: TCP/IP, DNS, DHCP, SNMP, ICMP, HTTP(S), REST APIs, SSL, TLS, VLANs, SMTP, POP3, IMAP, SMB, FTP, SSH
- Switching
  - Administration and configuration of Layer 2 and Layer 3 switches
  - VLAN implementation for network segmentation and improved security
  - Configuration of protocols such as STP (Spanning Tree Protocol), VTP (VLAN Trunking Protocol), and EtherChannel
  - Performance monitoring and control
- Server administration
  - Management of Linux and Windows Server: installation, configuration, and maintenance
  - Configuration of network services: DNS, DHCP, FTP, HTTP, HTTPS, SSH
  - User, permission, and security policy management
  - Virtualization and containers: VMware, VirtualBox, Docker
  - Resource monitoring with htop, top, netstat, iostat, ps
- Network device configuration
  - Configuration and management of routers, switches, and access points (Cisco, TP-Link)
  - Security policies with ACLs (Access Control Lists) and user authentication
  - Firmware updates and configuration backups to keep the network stable
  - Automation and monitoring tools such as Ansible, SNMP, Wireshark, and Nmap
- Structured cabling
  - Installation and maintenance of structured cabling systems
  - Familiarity with CAT5E, CAT6, CAT6A cables and the TIA/EIA-568 and ISO/IEC 11801 standards
  - Certification testing with specialized equipment (cable certifiers, continuity and attenuation testers)
  - Cabling organization and standardization for high performance and low latency
  - Fundamentals of fiber optics (FTTH, GPON, connectors and links)
- Observability
  - Studies and tests on logging systems with Syslog and Rsyslog

    - Interest in log collection, analysis, and visualization with tools such as the ELK Stack (Elasticsearch, Kibana)
    - Initial monitoring implementation with Zabbix, Nagios, Prometheus, and Grafana
    - Studies on dashboards for analyzing metrics and alerts on asset health and performance
    - Fault diagnosis and resolution based on logs, metrics, and distributed tracing
- Cloud: AWS, Azure, Google Workspace, Cloudflare, NGINX, Traefik
- Security: Firewall, WAF, HashiCorp Vault, Nomad
- Operating systems: NixOS, Linux as desktop (Ubuntu, Debian, Arch Linux, Fedora, NixOS, Mint), Linux as server (Ubuntu, Debian, CentOS), Windows desktop, Windows Server (2016, 2022), shell scripting

Certifications of Interest

- *CompTIA Network+*
- *AWS Certified Developer – Associate*
- *AWS Certified DevOps Engineer – Professional*
- *GitHub Actions Certified*
- *GitHub Advanced Security Certified*
- *Linux Foundation CKAD (Certified Kubernetes Application Developer)*
- *GitLab Certified CI/CD Specialist*

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